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Evaluating Parameter-Efficient Fine-Tuning vs. Prompt Engineering for Profile-Driven Recommendations

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Abstract

Large language models (LLMs) show significant promise for personalized recommendation systems, yet identifying the most effective adaptation strategy for structured data remains a challenge. This study compares parameter-efficient fine-tuning (PEFT) and in-context learning approaches for personalized item recommendation. We utilized explicit user profiles, encompassing specific likes and dislikes, alongside a comprehensive JSON-formatted item catalog. Two distinct adaptation paradigms were compared on Llama 3.1 8B and Llama 3.2 3B: PEFT, specifically Low-Rank Adaptation (LoRA) and Infused Adapter by Inhibiting and Amplifying Inner Activations (IA^3), against prompt engineering strategies optimizing input instructions without altering model weights. Because personalized recommendation is inherently subjective and has no objective ground truth, three human annotators independently judged 1,521 recommendations per configuration as agreeing (1), disagreeing (−1) or neither (0) with the recommendation, and the consistency of their judgments was quantified using Fleiss' kappa (κ). Agreement was almost perfect in all six configurations (κ between 0.863 and 0.904), indicating a reliable evaluation protocol. Fine-tuned adapters yielded higher agreement than the prompt-engineered baseline for both foundation models, although the differences between configurations were small (at most 0.041) and no single adapter type was consistently superior across models. These results show that adapter-based fine-tuning produces recommendations whose relevance to explicit user profiles is judged at least as consistently as that of prompt engineering alone.

Keywords: large language models; parameter-efficient fine-tuning; prompt engineering; recommender systems; Low-Rank Adaptation (LoRA); structured data processing; in-context learning

1. Introduction

The integration of Large Language Models (LLMs) into recommendation systems has fundamentally transformed the field, treating recommendation as a sequence generation task. However, adapting these models to dynamic user preferences presents significant challenges regarding stability and plasticity. To address the continuous evolution of users and items, recent research has explored parameter-efficient fine-tuning (PEFT) methods like Low-Rank Adaptation (LoRA); for instance, the PESO framework maintains a single evolving adapter anchored by a proximal term to dynamically capture new user behaviors while mitigating the catastrophic forgetting associated with sequential updates [1].

Received:
Revised:
Accepted:
Published:

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Transitioning LLMs into conversational recommenders introduces additional complexities, particularly concerning semantic misalignment and cumbersome training processes. To streamline this, the PECRS framework reformulates the problem entirely as a natural language processing task, utilizing a frozen pre-trained language model and rich textual metadata to perform both item recommendation and response generation. This unified architecture can be optimized in a single training stage using PEFT techniques, significantly reducing computational overhead [2].

While efficiency is crucial, standard PEFT methods often lack parameter-level personalization because they rely on a static, shared LoRA module for all users. The RecLoRA framework addresses this by introducing a Personalized LoRA module, which dynamically generates unique weights for each user through a soft routing mechanism guided by a conventional recommendation model [3].

Beyond parameter adaptation, prompt engineering and in-context learning offer alternative pathways for efficient LLM integration. Because routing all users through an LLM is computationally prohibitive, recent hybrid task allocation frameworks propose identifying "weak users" with sparse interaction histories and contextualizing their data into natural language instructions. This targeted approach drastically reduces token usage and latency while improving recommendation quality where traditional systems struggle [4].

Finally, applying these adaptations in complex, multi-domain industrial settings requires balancing general user behavior with domain-specific data sparsity. Frameworks such as MLoRA achieve this by combining a shared backbone, which learns generalizable information across domains, with specialized low-rank adaptive modules tailored to capture distinctive, domain-specific features [5].

2. Background and Related Work

The integration of Large Language Models (LLMs) into recommender systems has been systematically categorized by Elmoghazy et al., who established a taxonomy based on the LLM's specific role, learning paradigms, and overarching system architecture. Their comparative study emphasizes that while LLMs can act as textual feature encoders or direct recommendation generators, they face persistent challenges such as context window constraints, computational latency, and a disparity between language objectives and recommendation tasks [6]. Further structuring this domain, Bao et al. surveyed the field by dividing architectures into Discriminative and Generative paradigms. Their research details how generative models provide impressive zero-shot capabilities and natural language explainability, despite struggling with context length limits and inherent biases [7]. Additionally, Liu et al. explored language modeling paradigm adaptations, contrasting traditional pre-training with parameter-efficient prompt learning to overcome data sparsity, while noting ongoing issues with factual consistency and user privacy [8].

Expanding on these foundational taxonomies, Nawara and Kashef provided a structured examination of how LLM-powered systems overcome conventional recommendation limitations like data sparsity, cold-start issues, and long-tail challenges. Their survey categorizes recommenders into five distinct paradigms—discriminative, generative, hybrid, graph-enhanced, and multi-modal—while evaluating the transition from traditional metrics to modern generative assessments and "LLM-as-a-Judge" frameworks [9]. Similarly, Shehmir and Kashef conducted a comprehensive systematic literature review of over 150 works to analyze the integration of transformer-based backbones such as BERT, GPT, and T5 within recommendation contexts. Their findings map the application of prompt engineering and parameter-efficient fine-tuning (PEFT) across diverse applications while

highlighting persistent algorithmic hurdles like high inference latency and societal bias [10].

Shifting the focus specifically toward personalized environments, Chen et al. detailed how the integration of LLMs transforms user experiences from passive filtering into active, conversational, and explainable processes. They established a multifaceted taxonomy where LLMs act as content interpreters, system reasoners, conversational agents, and tool-learning controllers, though they noted significant operational constraints regarding industrial deployment costs and the "phantom" problem of hallucinated content [11]. To frame the technical necessity of tailoring these models, Ling et al. explored the mechanisms of domain specialization required to transform general-purpose LLMs into disruptive tools for highly specific fields. Their systematic framework categorizes adaptation techniques based on model accessibility into black-box external augmentation, grey-box prompt crafting, and white-box model fine-tuning, emphasizing that while fine-tuning offers deep alignment with domain objectives, it introduces risks of catastrophic forgetting and high resource overhead [12].

Transitioning from theoretical surveys to empirical frameworks, Xu et al. proposed LLMRec to systematically evaluate the performance of LLMs functioning directly as recommender systems. Their extensive empirical analysis demonstrated that fine-tuning open-source models via parameter-efficient methods like Low-Rank Adaptation (LoRA) yields superior domain adaptation compared to zero-shot approaches, although targeted reasoning prompts like Chain-of-Thought remain crucial for optimizing prediction accuracy [13].

Focusing specifically on the advancement of parameter-efficient methods, Zhu et al. addressed the inefficiency of static LoRA matrices by introducing RecLoRA. This framework dynamically aggregates multiple independent meta-LoRA weights to create a unique, personalized module for each user, successfully balancing recommendation effectiveness and processing time [3]. Building further on personalized adaptation, Kim et al. identified that standard PEFT foundation models often generate user-blind item embeddings. To resolve this, they proposed PerPEFT, a framework that groups users by similar interests and assigns a distinct, dedicated PEFT module to each demographic. By employing group-specific hard negative sampling, their architecture dynamically captures fine-grained item characteristics, proving highly adaptable to various structures like LoRA and IA^3 with minimal parameter overhead [14]. Addressing the dynamic nature of user profiles over time, Zeng et al. tackled the issue of shifting preferences and catastrophic forgetting associated with global fine-tuning. They introduced the Region-Aware Incremental Editing framework, which clusters historical interactions into distinct knowledge regions. When new behaviors emerge, a confidence-aware routing mechanism triggers localized updates to a specialized LoRA module dedicated solely to the affected region, effectively balancing the assimilation of novel interests with the retention of stable historical behaviors [15].

In domain-specific applications, Guo et al. utilized Quantized LoRA within the TIGMI framework to fine-tune an LLM for cultural tourism. This approach reduced computational overhead by 60 percent while successfully integrating Retrieval-Augmented Generation (RAG) for real-time multimodal interactions [16]. Similarly, Choi et al. applied LoRA to the upper layers of a BERT-based model to enhance a RAG framework for autonomous driving, effectively reducing trainable parameters and improving the retrieval of complex legal texts [17]. Furthermore, Gao et al. demonstrated the scalability of fine-tuned LLMs by developing a multi-dimensional framework to automatically assess the quality and reliability of massive volumes of tourist-generated contents across digital platforms [18].

Conversely, prompt engineering offers an effective alternative that completely avoids altering model weights. Nikolaychuk and Poddubniy applied prompt engineering to

the Saiga2 model to automatically extract and classify user sentiments from regional tourism data, successfully achieving high alignment with human expert evaluations [19]. To resolve the complex challenges of intent recognition and factual hallucinations, Huang et al. proposed a multi-layer prompt engineering architecture that decomposes user queries into sequential subtasks. By integrating this step-by-step reasoning strategy with real-time external APIs, their system drastically reduced hallucination rates, proving particularly effective for lightweight models in travel planning applications [20].

Based on the cumulative evidence from previous studies, the goal of this work is to examine how adapter-based PEFT compares with prompt engineering strategies for personalized item recommendation. Because personalized recommendation is inherently subjective and lacks an objective ground truth, the study assesses how consistently independent human annotators judge the recommendations produced under each adaptation approach, quantified using Fleiss' kappa. In particular, the study explores how different adaptation approaches, utilizing explicit user profiles and structured JSON-formatted catalogs, influence this agreement. Guided by prior literature, the following hypotheses are considered:

- H1: Annotator judgments of recommendations generated under every adaptation strategy may reach at least substantial agreement ($\kappa > 0.60$), indicating that the annotation protocol is reliable regardless of the adaptation method.
- H2: Recommendations generated by PEFT-adapted models may be judged more consistently (higher κ) than those generated by the prompt-engineered baseline of the same foundation model, as fine-tuning may align outputs more closely with explicit user preferences.
- H3: LoRA-based adapters may yield higher annotator agreement than other parameter-efficient techniques such as IA^3.
- H4: The relative ordering of adaptation strategies may be consistent across foundation models of different size and generation (Llama 3.1 8B and Llama 3.2 3B), indicating that the findings generalize beyond a single architecture.

3. Methodology

This section details the experimental framework designed to evaluate and compare the efficacy of parameter-efficient fine-tuning (PEFT) against prompt engineering for personalized point-of-interest (POI) recommendation. We outline the data collection process, the synthesis of user profiles for ground-truth annotation, the retrieval-augmented generation (RAG) pipeline, and the specific adaptation strategies applied to the foundation models.

3.1. System Architecture Overview

The proposed recommendation system is evaluated through a RAG framework tailored for structured POI data. The architecture deliberately isolates the retrieval mechanism to ensure a controlled, isolated comparison between two distinct Large Language Model (LLM) adaptation paradigms: Prompt Engineering and PEFT. Two foundation models—Llama 3.1 8B and Llama 3.2 3B—are evaluated to assess the scalability, robustness, and architectural generalizability of these adaptation strategies.

3.2. Data Collection and Vector Indexing

To simulate a real-world local recommendation engine, an exhaustive dataset of POIs within the city of Pula, Croatia, was curated via web scraping. The collected data spans multiple domains, including gastronomy, cultural heritage sites, and recreational areas.

The raw data was formalized into a standardized JSON schema. Each POI is represented as a single JSON object containing a categorical type and a nested properties

dictionary. To enrich the semantic density of the dataset and improve personalized retrieval, the following key metadata fields were populated:

- Taxonomy Tags: An array of descriptors (e.g., mediterranean, vegetarian) defining the core attributes of the POI.
- Binary and Categorical Constraints: Features such as outdoor_seating, indoor_seating, and price (e.g., low, medium, high) that serve as strict filtering criteria.
- Contextual Narratives: Comprehensive textual descriptions and up to three verified user reviews to provide the embedding model with nuanced sentiment and atmospheric context.

Each JSON object was treated as an independent document chunk. To enable dense retrieval, embeddings were generated using the nomic-embed-text:v1.5 model, selected for its strong performance on heterogeneous text structures. The resulting vectors were indexed using Facebook AI Similarity Search (FAISS) to facilitate efficient similarity computations.

3.3. Synthetic User Profile Generation and Ground Truth Annotation

Evaluating personalized recommendations necessitates a dataset that accurately models diverse user intents and constraints. We synthetically generated 5000 unique user profiles. Each profile represents a distinct permutation of demographic and behavioral traits, mapped to the JSON schema:

- Demographics: gender, age, and nationality to simulate cultural and generational variance.
- Behavioral Archetypes: travel_style (e.g., backpacker, luxury) and persona (e.g., pet_owner) to establish the user's primary motivation.
- Explicit Preferences: likes and dislikes structured as arrays of venue tags to serve as positive and negative alignment vectors.
- Quantitative Constraints: price_preference and min_rating to bound the acceptable POI thresholds.
- Natural Language Query (input): A conversational prompt synthesized from the profile's constraints (e.g., "Any nice outdoor spots where I can bring my pet?").

To establish a high-quality ground truth for training and evaluation, the Qwen 3.6 35B model was utilized to generate initial recommendation responses for all 5000 queries. Because personalized recommendation is inherently subjective, human annotators were employed to blindly review and validate the relevance of the generated recommendations against the explicit constraints of the user profiles.

To ensure the scientific rigor of the dataset, Fleiss' kappa (κ) was calculated to measure inter-rater reliability among the three human annotators. Fleiss' kappa was selected because it extends chance-corrected agreement to more than two raters, and its definition is given in Section 3.6. Only the subset of responses that achieved absolute consensus (i.e., unanimously marked as correct by all annotators) was retained for the PEFT training corpus. Responses exhibiting annotator disagreement or partial correctness were isolated and repurposed for the benchmark testing dataset, ensuring the models are evaluated on the most challenging, highly subjective edge cases.

3.4. Retrieval and Context Injection Pipeline

During inference, a standardized retrieval pipeline is maintained across all evaluated models to isolate the impact of the LLM adaptation strategy. For a given user input, the query is embedded and searched against the FAISS index.Initially, the vector database retrieves the top $K = 30$ candidate chunks based on vector distance. To maximize the signal-to-noise ratio within the LLM context window, these 30 candidates are subsequently

filtered, and only the top 10 most relevant JSON chunks are injected directly into the LLM prompt.

3.5. LLM Adaptation Strategies

The study contrasts a robust Prompt Engineering baseline against two state-of-the-art PEFT methodologies:

- Prompt Engineering (Baseline): The foundation models are augmented with extended, highly structured system instructions. This method relies entirely on in-context learning, leveraging the top 10 retrieved chunks without any underlying weight updates.
- Parameter-Efficient Fine-Tuning (PEFT): The models are fine-tuned on the consensus-driven training set using two distinct adapter mechanisms:
  - Low-Rank Adaptation (LoRA): Freezes the pre-trained model weights and injects trainable rank decomposition matrices into the transformer architecture, allowing the model to learn domain-specific ranking logic.
  - IA^3 (Infused Adapter by Inhibiting and Amplifying Inner Activations): A highly efficient alternative that scales inner activations using learned vectors, significantly reducing the number of trainable parameters and computational overhead compared to LoRA.

3.6. Evaluation Protocol and Inter-Rater Agreement

Six configurations were evaluated: Llama 3.1 8B and Llama 3.2 3B, each under prompt engineering, LoRA and IA^3. Each configuration generated recommendations for the same set of 1,521 user queries. For every recommendation, the three annotators independently assigned one of three labels: 1 (the annotator agrees with the recommendation), -1 (the annotator disagrees) or 0 (neither). Since no objective ground truth exists for subjective recommendation quality, the evaluation relies on the consistency of these independent judgments, quantified with Fleiss' kappa.

Let N be the number of evaluated recommendations, $n = 3$ the number of annotators, $k = 3$ the number of categories, and n_{ij} the number of annotators who assigned recommendation i to category j. The observed agreement for a single recommendation and its mean are

$$P_i = \frac{1}{n(n-1)} \left(\sum_{j=1}^{k} n_{ij}^2 - n \right), \qquad \bar{P} = \frac{1}{N} \sum_{i=1}^{N} P_i. \tag{1}$$

The expected chance agreement is computed from the overall proportion p_j of ratings in each category, and Fleiss' kappa follows as

$$p_j = \frac{1}{Nn} \sum_{i=1}^{N} n_{ij}, \qquad \bar{P}_e = \sum_{j=1}^{k} p_j^2, \qquad \kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}. \tag{2}$$

A value of $\kappa = 1$ indicates perfect agreement, while $\kappa = 0$ indicates agreement no better than chance. Values are interpreted using the scale of Landis and Koch [21]: slight (0.01–0.20), fair (0.21–0.40), moderate (0.41–0.60), substantial (0.61–0.80) and almost perfect (0.81–1.00) agreement [22]. Fleiss' kappa expresses how consistently annotators judge the recommendations, not whether the judgments are correct. Uncertainty was estimated by nonparametric bootstrap, resampling queries with replacement (4000 replicates) and reporting 95% percentile intervals. Differences between configurations were evaluated with a paired bootstrap on the same resampled queries.

4. Results and discussion

5. Limitations and Future Work

6. Conclusions

This section is not mandatory, but can be added to the manuscript if the discussion is unusually long or complex.

Author Contributions: Conceptualization, R.K., L.S. and S.B.Š.; methodology, L.S.; software, R.K.; validation, I.L. and S.B.Š.; formal analysis, R.K.; investigation, R.K. and L.S.; resources, R.K.; data curation, R.K. and L.S.; writing—original draft preparation, R.K. and L.S.; writing—review and editing, S.B.Š. and I.L.; visualization, R.K.; supervision, S.B.Š. and I.L.; project administration, S.B.Š. and I.L.; funding acquisition, I.L. All authors have read and agreed to the published version of the manuscript.

Funding: This work was (partially) supported by the EC Digital Europe Programme EDIH Adria 2.0 (101256325); SPIN projects IP.1.1.03.0120, IP.1.1.03.0158 and IP.1.1.03.0039; and NextGenerationEU University grants: uniri-iz-25-6, uniri-iz-25-220, IIP_010144, IIP_010136

Data Availability Statement: The original data presented in this study are openly available in a public repository at https://www.mdpi.com/ethics.

Acknowledgments: This research was performed using the Advanced computing service provided by University of Zagreb University Computing Centre - SRCE.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

LLMs	Large Language Models
NLP	Natural Language Processing
PEFT	Parameter-Efficient Fine-Tuning
LoRA	Low-Rank Adaptation
IA^3	Infused Adapter by Inhibiting and Amplifying Inner Activations
ICL	In-Context Learning
JSON	JavaScript Object Notation
SFT	Supervised Fine-Tuning
GPU	Graphics Processing Unit
DDP	Distributed Data Parallel
MLP	Multi-Layer Perceptron
bf16	BFloat16 Precision Format
tf32	TensorFloat-32

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